471

Accelerated Feature and Pose Estimation for Time-Critical Pick-and-Place Applications

Rajat Kumar Thakur, Isha Jangir, Siddharth Nimbalkar, Deepika Gupta, and Jignesh Patel
Indian Institute of Information Technology Vadodara International Campus Diu, India

Abstract

In warehouse automation, managing unordered piles of objects poses a significant challenge. This paper introduces a novel method to detect geometric features like edges and corners in unstructured 3D point clouds for robotic tasks.

The approach uses an eigenvalue-based surface variation measure to quickly extract sharp edge points from raw data, outperforming traditional methods in speed and efficiency. A 3D Harris corner detector identifies key corners, enabling reliable pose estimation for texture-less objects.

This makes it ideal for real-time pick-and-place operations, enhancing autonomous grasping in cluttered environments and advancing automation in construction and manufacturing.

Motivation

Robotic pick-and-place in cluttered warehouses is hindered by slow geometric methods and deep learning approaches that fail on texture-less objects. A fast, and robust solution is critically needed for real-time automation.

Introduction

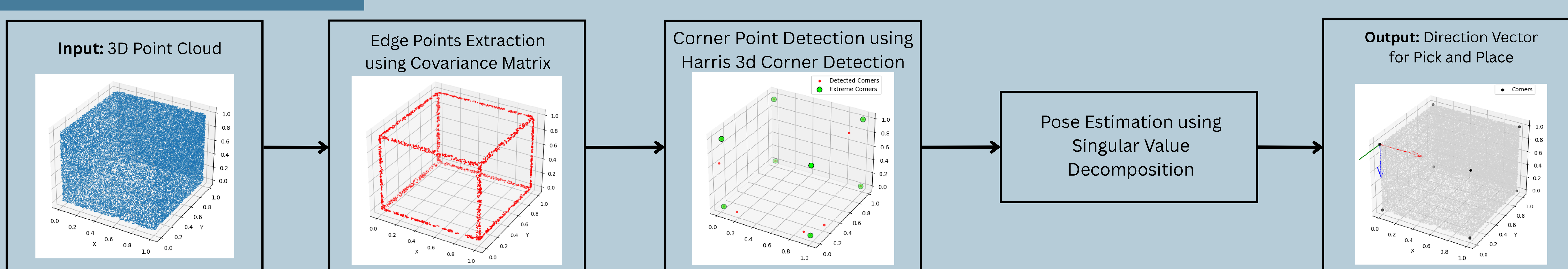
We present a fast geometric pipeline for real-time pose estimation of texture-less objects in cluttered warehouses.

By combining eigenvalue-based edge extraction, 3D Harris corner detection, and closed-form SVD alignment from extreme corners, our method delivers accurate features and pose with 3.3× speedup over the state-of-the-art baseline.

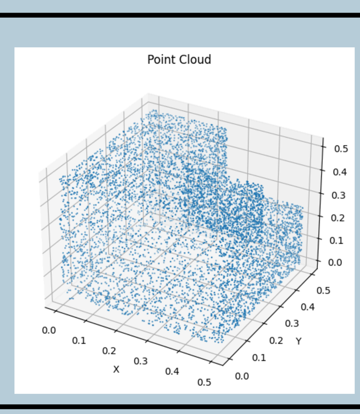
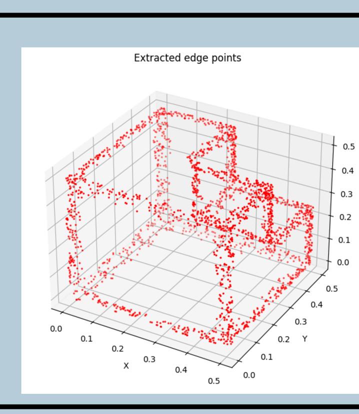
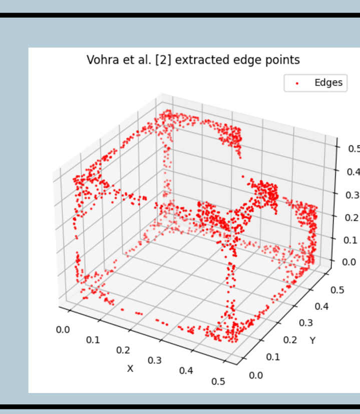
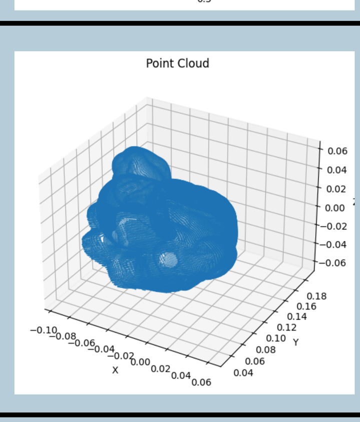
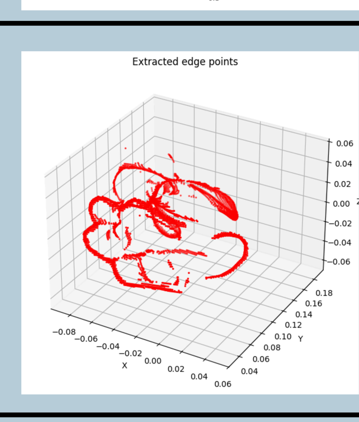
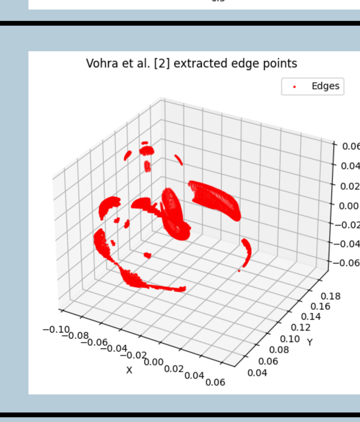
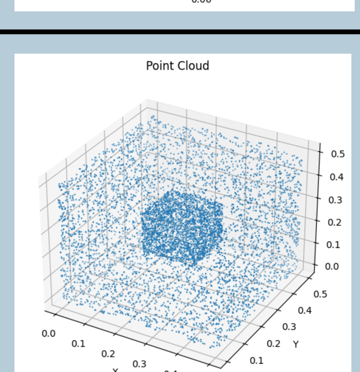
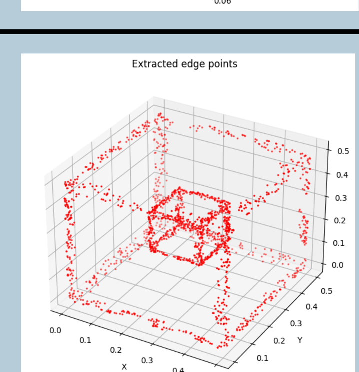
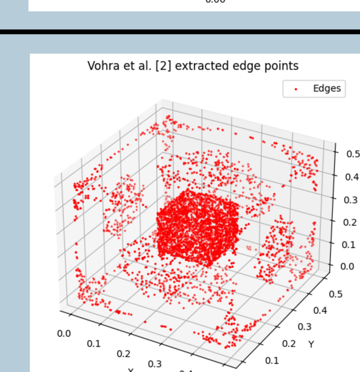
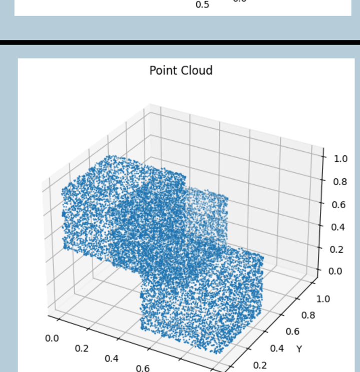
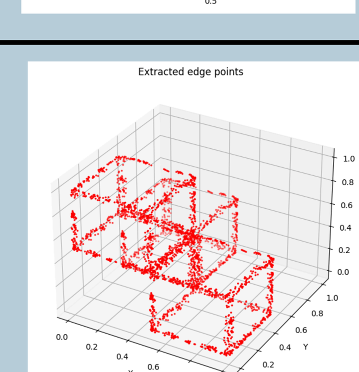
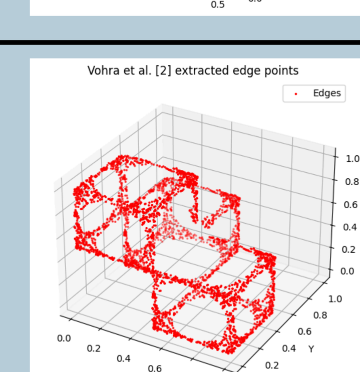
Novel Contributions

- **Integrated Feature Pipeline:** Our closely integrated pipeline combines eigenvalue-based surface variation for rapid edge extraction with 3D Harris corner detection, yielding robust geometric features for texture-less objects in unstructured point clouds with minimal parameter tuning requirements.
- **Efficient Pose Estimation:** We enable corner-driven pose estimation using only observed extreme corners and closed-form SVD alignment, generating reliable hypotheses for real-time pick-and-place.
- **Computational Efficiency:** We Demonstrated 3.3× runtime speedup relative to recent multi-stage geometric baselines, with comprehensive analysis identifying optimization opportunities for future enhancements.

Proposed Methodology



Qualitative comparison of edge extraction results

Original Point Cloud	Eigenvalue based edge detection	Vohra et al.'s method.
		
		
		
		

Experimental Results

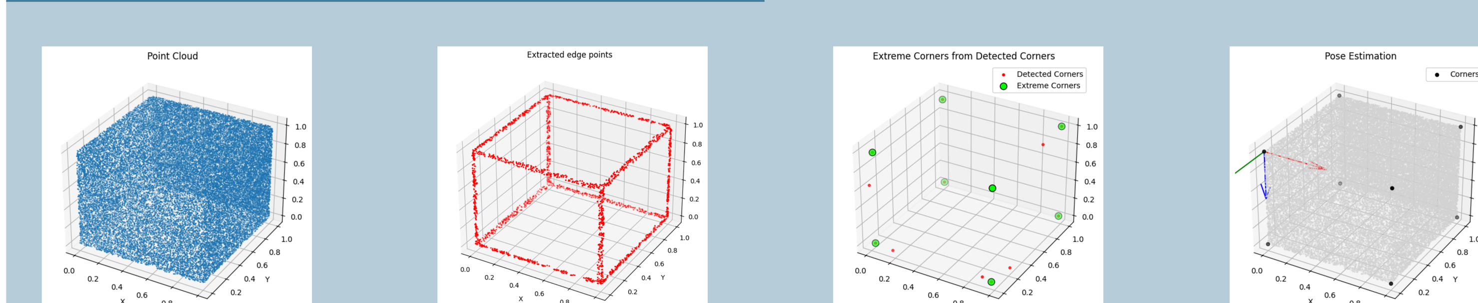


Fig. Our pipeline on synthetic cube point cloud

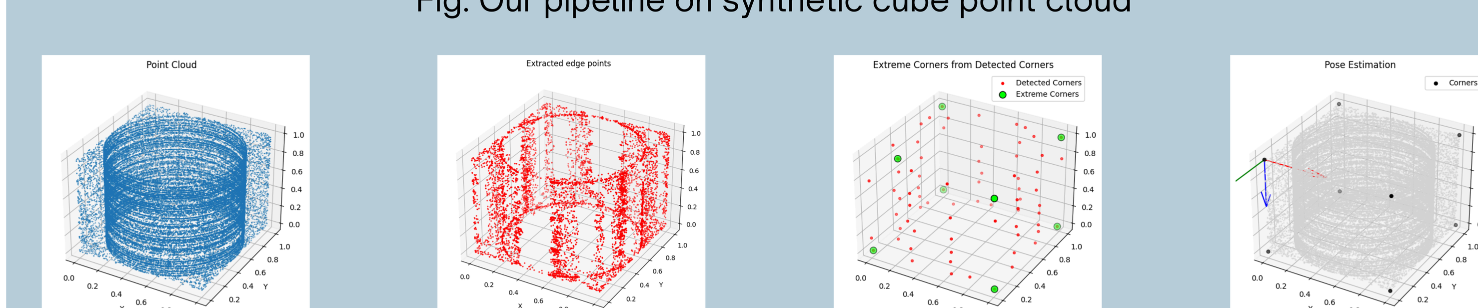


Fig. Our pipeline on synthetic cylinder inside a hollow cube point cloud data

Model	Edge	Corner	Pose Est.	Total Time
Our Methodology	5.520	0.573	1.850	7.943
Vohra et al	24.168	1.110	1.551	26.829

Table 1: Computation time (sec) at each step

Conclusion

This work presents an efficient, robust framework for edge, corner, and pose estimation in unstructured 3D point clouds, optimized for real-time robotic pick-and-place.
Integrating eigenvalue-based edge extraction, 3D Harris corner detection, and corner-driven pose estimation, it achieves precise feature identification with minimal tuning and 3.3× speedup over baselines.
Overall, this framework lays a fast, reliable, and extensible foundation for robotic manipulation in real-world environments.

References

[1] Vohra, M., Prakash, R., Behera, L.: Edge and corner detection in unorganized point clouds for robotic pick and place applications. arXiv:2104.09099 [cs.RO] (2021)
[2] Bazazian, D., Casas, J.R., Ruiz Hidalgo, J.: Fast and robust edge extraction in un organized point clouds. In: Proc. 2015 Int. Conf. Digital Image Computing (2015)